



PRODUCT NAME

Premium Face Whitening Cream Formula

Description

A premium face cream formulated to deliver skin whitening effects, even skin tone, intense moisturizing, UV protection, and gentleness on sensitive skin.

Technical Specifications

pH Level:	7
Viscosity:	High
Shelf Life:	24 months
Batch Size:	1,000 - 5,000 units
Processing Time:	Approximately 2 hours
Temperature:	Processing temperature initially at 75°C, cooling to room temperature
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Large mixing vessel, heating mantle, homogenizer, standard laboratory glassware
Certification:	Must comply with current EU Cosmetics Regulation (EC) No. 1223/2009

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	50.0%	Solvent
2	Niacinamide	Niacinamide	5.0%	Skin whitening and even tone
3	Glycerin	Glycerin	10.0%	Moisturizing
4	Octyl Methoxycinnamate	Ethylhexyl Methoxycinnamate	6.0%	UV protection
5	Cetearyl Alcohol	Cetearyl Alcohol	5.0%	Emulsifier and thickening agent
6	Ceteareth-20	Ceteareth-20	4.0%	Emulsifier
7	Caprylic/Capric Triglyceride	Caprylic/Capric Triglyceride	10.0%	Emollient
8	Ascorbyl Palmitate	Ascorbyl Palmitate	5.0%	Antioxidant for skin brightening
9	Phenoxyethanol	Phenoxyethanol	0.9%	Preservative
10	Disodium EDTA	Disodium EDTA	0.1%	Stabilizer
11	Fragrance	Parfum	1.0%	Add pleasant scent
12	Titanium Dioxide	Titanium Dioxide	3.0%	Physical UV filter
Total			100%	

Manufacturing Process

Phase 1: Phase A

- Combine Water, Niacinamide, and Glycerin in a large vessel.
- Heat the mixture to 75°C under continuous stirring until homogeneous.

Phase 2: Phase B

1. In a separate vessel, melt Cetearyl Alcohol and Cetearth-20 until fully unified.
2. Add Caprylic/Capric Triglyceride and Ascorbyl Palmitate to the melted Phase B ingredients.
3. Heat to 75°C while stirring.

Phase 3: Phase C

1. Combine Phase A and Phase B by slowly adding Phase B to Phase A with vigorous mixing.
2. Maintain the temperature at 75°C and continue mixing for 20 minutes.

Phase 4: Phase D

1. Cool the combined mixture to 40°C, then add Octyl Methoxycinnamate and Titanium Dioxide while stirring gently.
2. Add Phenoxyethanol, Disodium EDTA, and Fragrance under continuous mixing until fully incorporated.

Required Equipment

Large mixing vessel, heating mantle, homogenizer, standard laboratory glassware

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Must comply with current EU Cosmetics Regulation (EC) No. 1223/2009

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 1,000 - 5,000 units.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of the cream to clean facial skin each morning. Use a gentle upward motion until fully absorbed. Suitable for daily use.